

Recovering Psilocybin-Related Working-Memory Signatures by Perturbing a Recurrent Neural Network

Abstract

Acute psilocybin produces heterogeneous effects across human working-memory and attention tasks, including response slowing, greater impairment at higher updating load, and vulnerability to competing information. This dissertation tests whether such a selective behavioural signature can be recovered by perturbing a recurrent neural network (RNN) trained to perform working-memory tasks. We first established a fixation-gated continuous-time RNN that maintained a circular stimulus through variable delays, replicated its performance across five independently trained models, and evaluated independent Gaussian hidden-state noise as a non-specific degradation control. Mean response error was $4.36 \pm 1.55^\circ$ and fixation accuracy was 0.9677 ± 0.0016 across models; hidden-state decoding recovered delay-period angle with $0.55 \pm 0.46^\circ$ error. Gaussian noise caused graded, delay-dependent impairment while largely preserving fixation and circular centroid spacing. These results establish a reproducible baseline and quantify generic stochastic disruption. Gaussian noise caused broad, delay-dependent degradation rather than recovering a selective human behavioural signature. It therefore provides a control for evaluating literature-motivated perturbations, without being interpreted as a biological model of psilocybin.

1 Introduction

Working memory depends on retaining task-relevant information after its eliciting input has disappeared. Recurrent dynamics provide a natural computational substrate for this function: activity can settle onto an attractor or evolve along a slow manifold that preserves a decodable memory variable (Wang et al., 2013; Kilpatrick et al., 2013; Ghazizadeh and Ching, 2021). For a continuous circular variable, the relevant question is not only whether a network produces an accurate response, but whether its hidden population maintains ordered geometry, resists drift, and generalises beyond the delays encountered during training.

The dissertation’s central question is whether perturbing this recurrent system can reproduce the qualitative pattern of working-memory effects observed in humans during acute psilocybin administration. Human findings are mixed rather than uniformly impairing. Psilocybin has produced greater impairment under higher N-back updating load

(Barrett et al., 2018), while a small study found clearer impairment in multiple-object tracking than in spatial working-memory span (Carter et al., 2005). The behavioural target is therefore a selective signature—including slowing, load sensitivity, and susceptibility to competing information—rather than an arbitrary reduction in accuracy.

Psychedelic neuroscience motivates candidate dynamical explanations for this signature. The Entropic Brain hypothesis proposes that psychedelic states occupy a broader, less constrained repertoire of brain states (Carhart-Harris et al., 2014). Human MEG, EEG, and fMRI studies have reported increased signal diversity or altered network entropy under psilocybin and related compounds (Schartner et al., 2017; Timmermann et al., 2019; Viol et al., 2017). These measurements are direct evidence about recorded brain signals, but they do not imply that psilocybin is equivalent to additive noise in a task-trained RNN. Increased diversity could reflect structured changes in responsiveness, coupling, context dependence, or state-space exploration.

REBUS (“relaxed beliefs under psychedelics”) places this change within predictive processing (Carhart-Harris and Friston, 2019). The theory proposes reduced precision of high-level priors and greater influence of bottom-up signals, relating precision to post-synaptic gain. A working-memory translation is plausible because a maintained representation must resist irrelevant input. However, scaling an RNN’s input drive, recurrent drive, or activation slope is an implementation hypothesis; REBUS does not prescribe any of these equations. The distinction matters because the three manipulations can make different predictions about clean maintenance, distractor capture, and recovery.

A more direct computational precedent comes from the receptor-informed whole-brain model of Herzog et al. (2023). That study represented 5-HT_{2A} receptor activation as regionally heterogeneous modulation of the frequency–current response function. Response gain interacted with structural connectivity to produce non-uniform entropy changes. The study therefore supports response-gain modulation as a candidate modelling dimension and suggests that recurrent topology shapes its consequences. It did not study working memory, a trained RNN, or a pharmacological mapping for the abstract gain values used here.

The direction and population specificity of gain changes cannot be inferred from that whole-brain result alone. In mouse visual cortex, the hallucinogenic 5-HT_{2A} agonist DOI reduced visually evoked response gain (Michaël et al., 2019); cell-type-specific manipulation of a 5-HT_{2A}-related pathway produced heterogeneous effects and divisive suppression through balanced excitation and inhibition (Barzan et al., 2024). Separately, human EEG modelling linked perceptual precision to the synaptic gain of superficial pyramidal prediction-error units (Brown and Friston, 2012). These studies support bidirectional, pathway-specific tests, not a presumption that a psychedelic-like manipulation is a uniform gain increase.

We hypothesise that a literature-motivated perturbation of a trained working-memory RNN can recover the qualitative human signature more closely than generic stochastic disruption. This requires more than decreasing performance: a successful perturbation should reproduce a selective pattern across clean maintenance, delay duration, response

dynamics, updating demand, and competing input. The present analysis establishes the unperturbed baseline and tests independent Gaussian noise as the generic control. Gain modulation is retained as a candidate mechanism because it can alter response sensitivity or the balance of external and recurrent drive, but any such manipulation is an abstract computational hypothesis rather than a biological equivalent of psilocybin.

2 Methods

2.1 Fixation-gated circular task

Each stimulus angle $\theta \in [0, 2\pi)$ was encoded over 32 evenly spaced preferred angles μ_j using

$$c_j(\theta) = \exp\{\kappa[\cos(\theta - \mu_j) - 1]\}, \quad \kappa = 8. \quad (1)$$

Trials comprised pre-cue fixation, cue, delay, and response periods. Pre-cue duration was sampled from 15, 25, or 35 steps; cue duration from 10, 20, or 30 steps; and delay duration from 10, 20, 40, or 80 steps. The response lasted 25 steps. A fixation input was one until response onset and zero thereafter. The target fixation output followed the same schedule; the 32 circular outputs were silent before response and reported the remembered angle during response. Thus delay-period memory was evaluated from hidden state rather than from the deliberately silent circular readout.

2.2 Network and implemented recurrence

The model contained 33 inputs, 64 recurrent units, and 33 outputs. With $\alpha = dt/\tau = 20/100 = 0.2$, the code implemented

$$z_t = W_{\text{in}}x_t + b_{\text{in}} + W_{\text{rec}}h_{t-1} + b_{\text{rec}}, \quad (2)$$

$$h_t = \tanh[(1 - \alpha)h_{t-1} + \alpha z_t + \varepsilon_t], \quad (3)$$

$$y_t = W_{\text{out}}h_t + b_{\text{out}}. \quad (4)$$

This placement of the leak, perturbation, and non-linearity is the implemented update; it should not be replaced by a textbook continuous-time equation. During training only, independent recurrent noise with standard deviation 0.05 was added after the leaky update and before tanh.

2.3 Training and replication

We trained with Adam (learning rate 10^{-3}) for 4,000 steps using batches of 64 trials. Gaussian input noise had standard deviation 0.01. The weighted mean-squared error assigned weight 5 to circular response output and weight 2 to fixation output. The first five trial steps and the first five response transition steps were unscored. Gradients were clipped to a maximum norm/value of 1.0 as implemented by the training utility. Five

networks were trained from independent seeds 20260714–20260718. Model seed, rather than trial or stochastic batch, was the replication unit.

2.4 Baseline evaluation and dynamical analyses

Each network was evaluated over 20 batches. Response angle was decoded by population-vector averaging, and absolute circular error was the shortest wrapped angular distance from target. Fixation accuracy measured correct wait-go output state. Generalisation sweeps used delays of 10, 20, 40, 80, and 160 steps.

Delay-period content was decoded from hidden states by ridge regression to the two-dimensional target vector

$$[\cos(\theta), \sin(\theta)].$$

The decoder used 512 trials, a 75:25 train-test split, and ridge coefficient 1.0. Cross-temporal matrices fitted a decoder at each training time and tested it at each held-out time.

For the canonical checkpoint only, late-delay states seeded blank-input autonomous probes and fixed-point optimisation. Local Jacobians were computed at the resulting approximate fixed points. We report residual, decoded angle, drift, eigenvalue magnitude, and alignment of the leading eigenvector with the sampled circular tangent. These sampled analyses are descriptive properties of one checkpoint, not five-model estimates or proof of a global attractor.

2.5 Independent Gaussian control

Frozen networks received perturbations only during the delay. For each active time t , trial b , and hidden unit i , an independent $\eta_{tbi} \sim \mathcal{N}(0, 1)$ was drawn and normalised as

$$\varepsilon_{tbi} = \sigma \frac{\eta_{tbi}}{\left(|\mathcal{A}|^{-1} \sum_{(t,b,i) \in \mathcal{A}} \eta_{tbi}^2\right)^{1/2}}, \quad (5)$$

where $\mathcal{A}$ contains all delay-active entries. The realised within-batch RMS therefore equalled target strength σ . We evaluated delays 20, 80, and 160 at

$$\sigma \in \{0, 0.01, 0.025, 0.05, 0.10\}.$$

Each model-delay condition used five stochastic replicates of 20 paired 64-trial batches. Clean-fitted decoders quantified representational distortion without adapting to the noisy states. RMS is an intervention magnitude, not a pharmacological dose.

Outcomes were response error, clean-decoder error, angular drift, fixation accuracy, circular-state geometry, and response settling. Geometry was measured in a clean-fitted principal-component plane using within-angle dispersion and between-angle centroid spacing. Settling was the first scored response step with error below 20° after excluding the five transition steps.

3 Results

3.1 The baseline replicated accurate circular working memory

Across five independently trained models, mean response angular error was $4.36 \pm 1.55^\circ$ (range $2.95\text{--}7.03^\circ$), response population MSE was 0.00957 ± 0.00194 , and fixation accuracy was 0.9677 ± 0.0016 (mean $\pm$ SD across model seeds). Delay-period diagonal decoding error was $0.55 \pm 0.46^\circ$ (range $0.31\text{--}1.37^\circ$), showing that the remembered angle remained linearly recoverable despite the silent circular output.

The five models generalised to the untrained 160-step delay, with seed-level mean response errors ranging from 5.14° to 11.73° (Fig. 1). This increase indicates accumulating delay-dependent error rather than perfect time invariance.

3.2 A canonical checkpoint showed sampled attractor-like geometry

For the canonical checkpoint, fixed-point optimisation produced a mean one-step residual of 0.00202, with 28.1% of sampled endpoints below the pre-specified 0.001 residual threshold. Mean decoded fixed-point error was 1.24° and mean drift from the corresponding late-delay state was 1.10° . The mean leading eigenvalue magnitude was 0.99975, the mean second-largest magnitude was 0.88742, and leading-eigenvector alignment with the sampled circular tangent was 0.978.

Together, preserved angle, low drift, and a near-neutral tangent direction support an attractor-like or slow-manifold account (Fig. 3). They remain descriptive because the analysis used one checkpoint and sampled initial states.

3.3 Gaussian noise caused weak, delay-dependent degradation

Gaussian noise produced graded impairment (Fig. 4). At RMS 0.05, response error increased from $4.34 \pm 1.54^\circ$ to $4.91 \pm 1.45^\circ$ at delay 20, from $6.49 \pm 1.93^\circ$ to $7.47 \pm 1.68^\circ$ at delay 80, and from $9.03 \pm 2.50^\circ$ to $10.44 \pm 2.24^\circ$ at delay 160. At RMS 0.10, the corresponding errors were $6.42 \pm 1.35^\circ$, $10.48 \pm 1.92^\circ$, and $14.68 \pm 2.57^\circ$.

Maintenance measures were more sensitive than final response. At delay 80 and RMS 0.05, clean-decoder error increased from $1.29 \pm 0.82^\circ$ to $3.71 \pm 1.53^\circ$ and decoded drift from $2.60 \pm 1.84^\circ$ to $5.20 \pm 2.11^\circ$. Fixation accuracy remained 0.98 ± 0.00 in both conditions (Fig. 5).

At delay 80 and RMS 0.05, within-angle dispersion increased from 0.550 to 0.732, whereas normalised between-angle centroid spacing remained 0.990 relative to the clean value of one (Fig. 6). The representation therefore blurred without a wholesale collapse of its circular organisation.

Response settling changed little at the primary RMS. At delay 80, the first response step below 20° increased from 5.02 clean to 5.19 at RMS 0.05, with 99.2% of trials settling.

At RMS 0.10, settling increased to 6.55 steps and 93.6% settled (Fig. 7). Final error in this focused analysis was 5.94° clean, 6.75° at RMS 0.05, and 10.40° at RMS 0.10.

Independent Gaussian noise is therefore a useful weak, non-specific degradation control. It increases dispersion, drift, and error with elapsed delay, but does not identify a pharmacological or cognitive mechanism. Structured AR(1) and topology-correlated noise remain a separate exploratory robustness study and are not part of the present mechanism comparison.

4 Discussion

The fixation-gated RNN provides a replicated continuous working-memory baseline. Accurate response and delay-period decoding across five seeds show that the result is not restricted to one favourable initialisation. The canonical checkpoint’s sampled fixed points and local Jacobians are consistent with a ring-like slow manifold, although that inference is descriptive and does not substitute for replicated dynamical analysis.

The dissertation asks whether perturbing a working-memory RNN can recover the selective behavioural signature observed in humans under acute psilocybin. The present results do not yet answer that hypothesis, because they establish the baseline and generic control rather than a successful psilocybin-motivated manipulation. They nevertheless provide the necessary reference against which recovery of the human signature can be judged.

Independent Gaussian perturbation supplied the intended generic control. Its effects were graded and strongest in maintenance-sensitive measures, while fixation and between-angle organisation were largely preserved at moderate strength. This establishes how unstructured variability degrades the trained system. It does not establish that altered entropy in psychedelic recordings is Gaussian, nor that RMS maps to human dose. More importantly, the principal pattern was a broad increase in error with perturbation strength and delay. This is insufficient by itself to recover the task-selective combination of slowing, updating-load sensitivity, and competing-input vulnerability that defines the dissertation hypothesis.

The gain literature supplies one possible mechanistic interpretation, but not a completed result. Receptor-informed modelling supports altered neural response gain (Herzog et al., 2023), whereas REBUS motivates altered precision and effective afferent influence (Carhart-Harris and Friston, 2019). Direct sensory-cortex studies also caution that gain effects can be reduced, heterogeneous, and population specific (Michaie et al., 2019; Barzan et al., 2024). Consequently, any later RNN gain manipulation must be judged by whether it recovers the human behavioural pattern, not by whether it simply changes network activity or increases error.

5 Conclusion

The fixation-gated RNN provides a reproducible substrate for testing the dissertation’s central hypothesis. Independent Gaussian noise produces predictable delay-dependent degradation, but the current evidence does not show that it recovers the selective behavioural signatures reported in humans under acute psilocybin. The relevant standard for subsequent perturbations is therefore pattern recovery across working-memory conditions, not impairment alone. Even if an RNN perturbation meets that standard, it would demonstrate computational sufficiency rather than biological equivalence to psilocybin.

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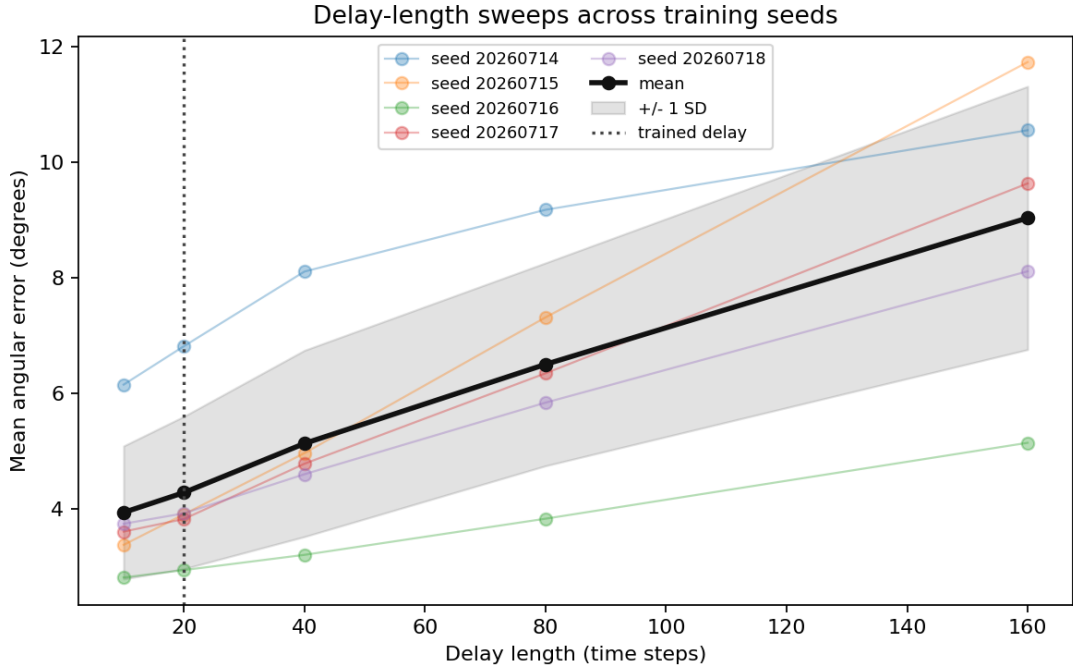


Figure 1: **Baseline delay generalisation across five independently trained networks.** Response angular error is shown for trained delays 10–80 steps and the untrained 160-step delay. Curves identify model seeds, which are the replication unit. Performance remains substantially above a random circular response at long delays, while between-seed variation increases.

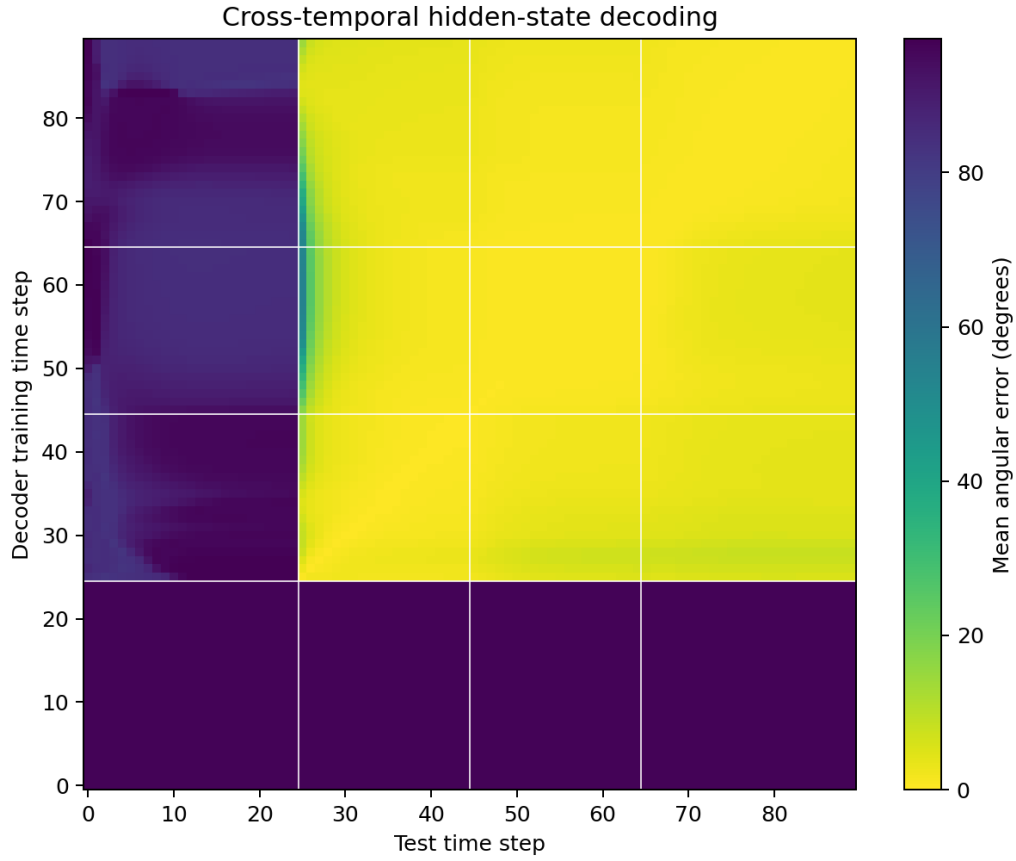


Figure 2: **Cross-temporal hidden-state decoding for the canonical checkpoint.** Ridge decoders predict sine and cosine of target angle from held-out hidden states. The low delay-period diagonal error (0.427° ; maximum 0.474°) confirms precise maintenance; the full matrix describes how the representational code transfers across time.

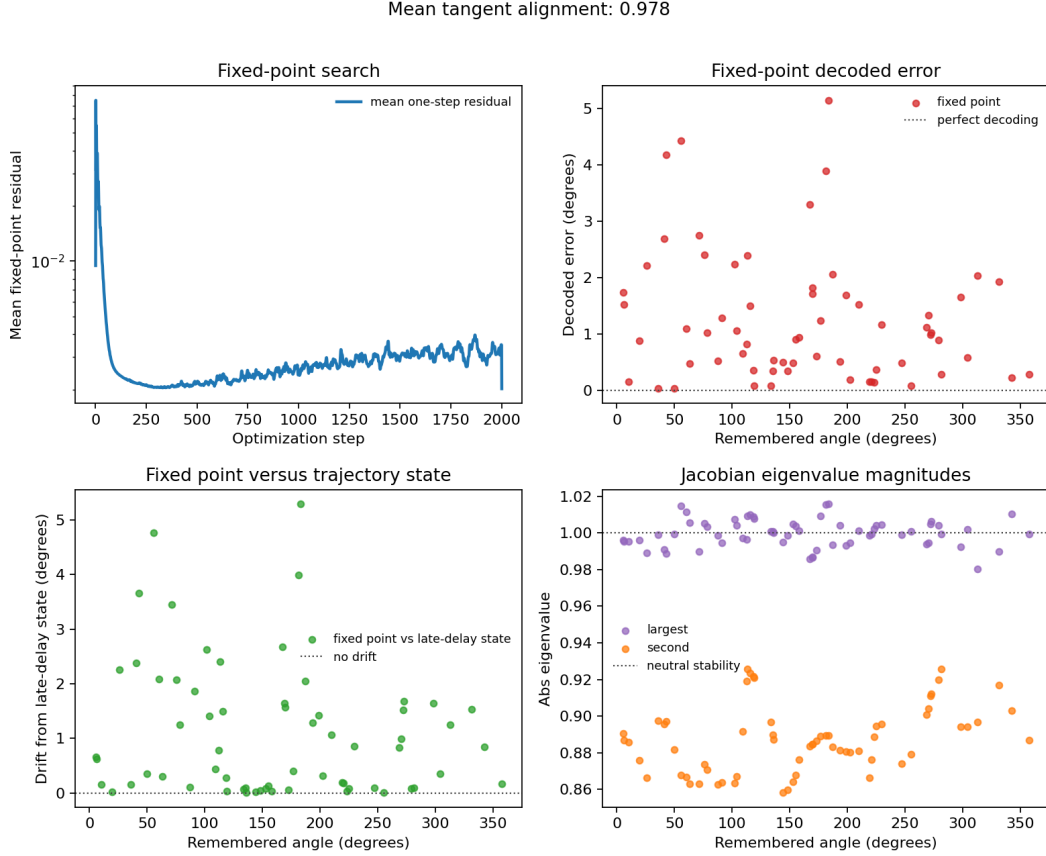


Figure 3: **Sampled fixed-point and Jacobian analysis of one canonical check-point.** Optimisation residuals, hidden-decoder angle preservation, drift, and local eigenvalue magnitudes are shown for late-delay starts. A near-neutral leading direction with contracting secondary dynamics is consistent with a ring-like slow manifold. The finite sample and incomplete convergence preclude an exhaustive claim about the full state space.

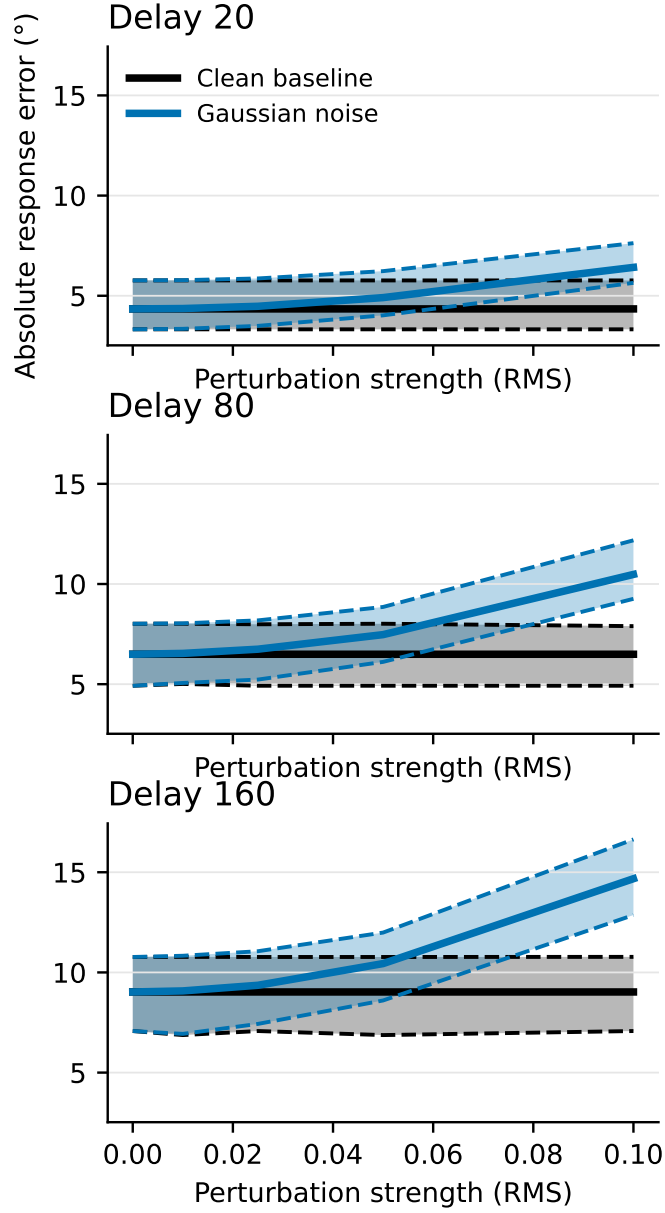


Figure 4: **Response error under independent Gaussian delay perturbation.** Absolute circular error is shown across RMS strengths and delay lengths. Solid lines are five-model means; dashed lines and shading are 95% bootstrap intervals across models. RMS quantifies perturbation magnitude, not drug dose.

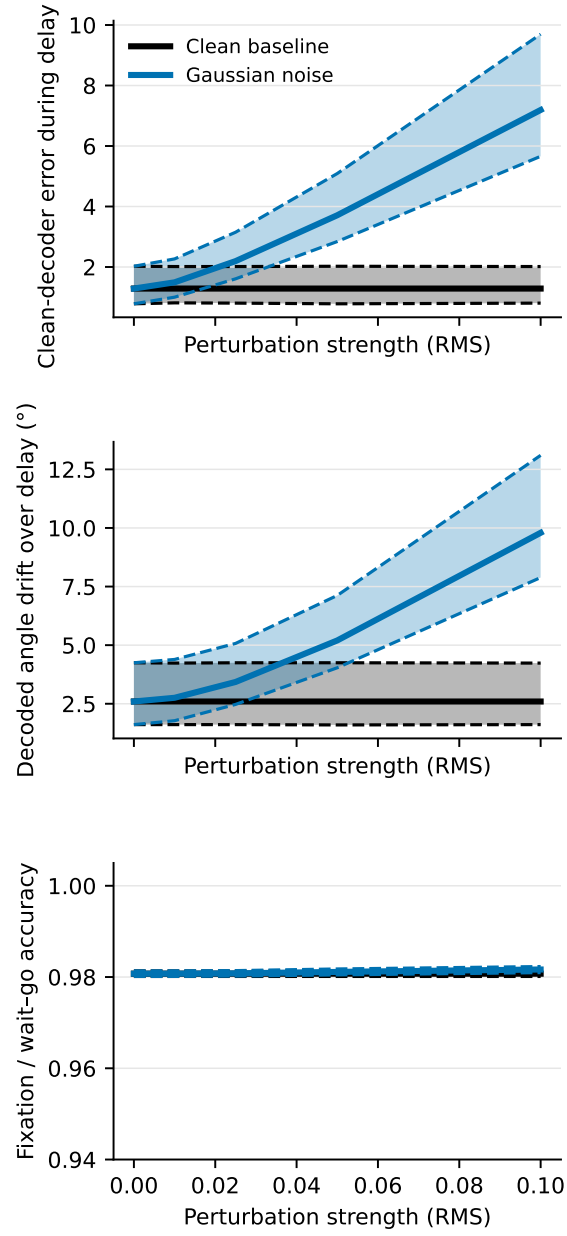


Figure 5: **Maintenance outcomes at delay 80.** Clean-fitted decoder error and decoded drift increase with perturbation RMS, whereas fixation accuracy is preserved. Lines and intervals are defined as in Fig. 4.

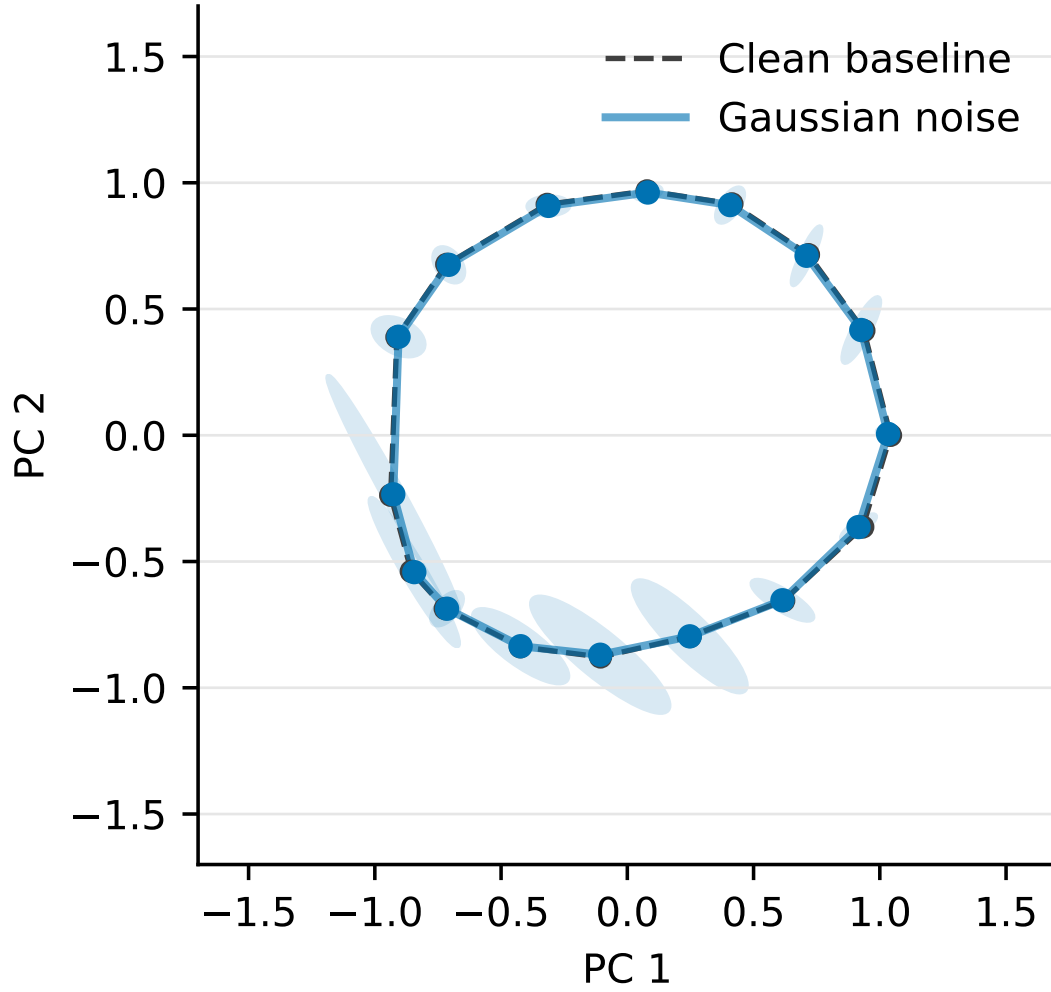


Figure 6: **Clean and Gaussian circular-state geometry at delay 80 and RMS 0.05.** Angle centroids and uncertainty ellipses are projected into the clean-fitted PC1–PC2 plane. Gaussian perturbation broadens within-angle states while largely preserving global centroid separation.

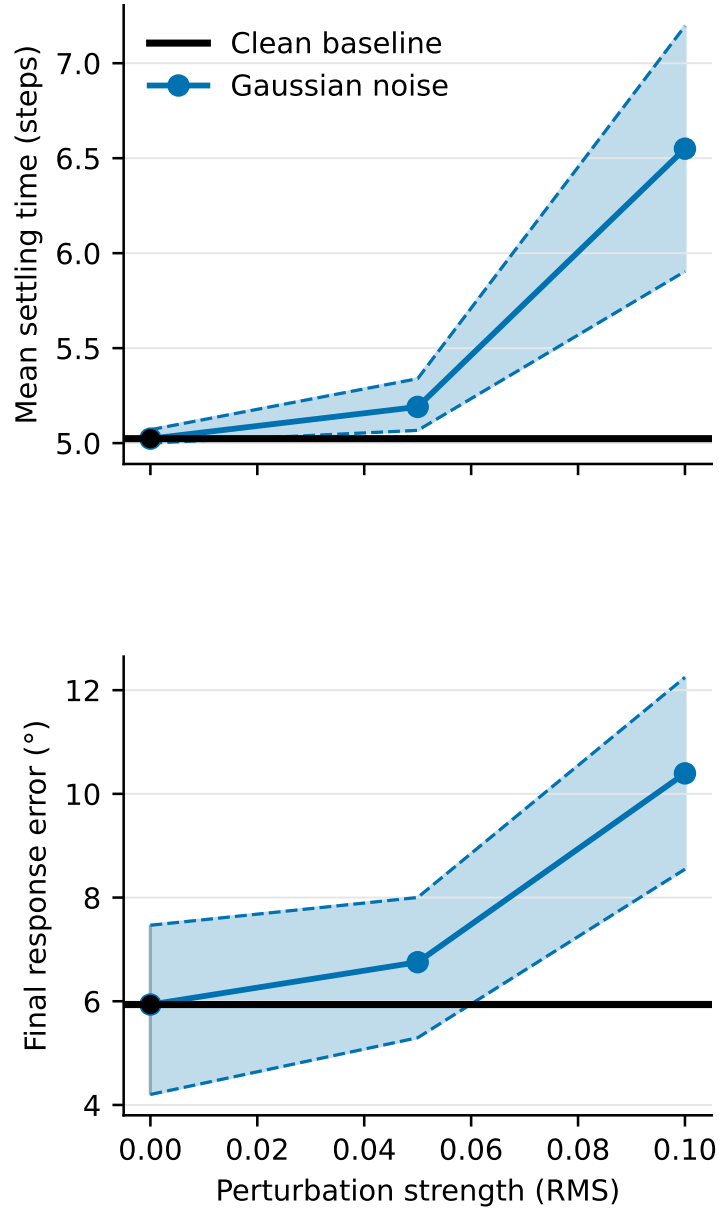


Figure 7: **Response settling at delay 80.** Settling is the first counted response step below 20° error after excluding the five transition steps. Points are model seeds and lines are five-model means.